

WEEK 6 PROJECT REPORT

HEALTHCARE VISUALIZATION & DASHBOARD

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Project Title: Healthcare Data Analysis using Power BI

Tool Used: Microsoft Power BI Desktop

Dataset: Healthcare Dataset

Introduction

The Week 6 Power BI project focuses on **Healthcare Visualization and Dashboard Development**. The objective of this task was to transform the healthcare data model created in the previous stage into an interactive and informative Power BI dashboard.

Different visualization types were used to present **patient demographics, medical conditions, admission volumes, billing information, hospital-related analysis and admission trends**. Interactive slicers were also added to allow users to dynamically filter and explore the dashboard.

Previous Week Workflow

In Week 5, the **Healthcare Dataset** was cleaned and transformed using Power Query. An **Age Group** column was created, Admission Types were standardized, and **Merge Queries** was used to combine the required information. The data was then structured into **Dim_Patient, Dim_Admission and Fact_Billing** tables with appropriate relationships in a star-schema model. Finally, DAX measures such as **Total Patients, Total Billing Amount, Average Billing Amount, Average Length of Stay and Readmission Rate** were created. This prepared the healthcare data model for the visualizations and dashboard developed in Week 6.

Objectives

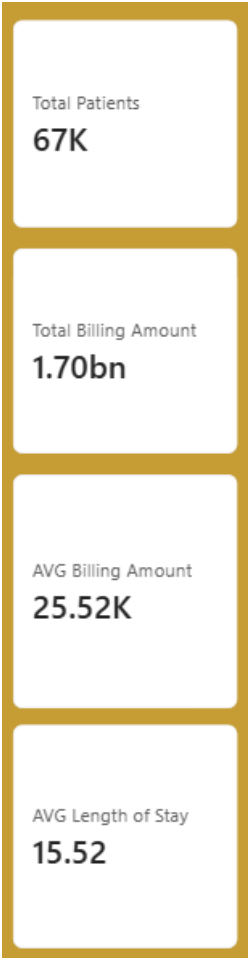
The main objectives of Week 6 were:

- Create an interactive healthcare dashboard in Power BI.
- Display important healthcare KPIs using summary cards.
- Analyze patient demographics and distribution.
- Analyze medical conditions and admission volumes.
- Visualize admission trends over time.
- Analyze billing amounts by insurance provider.
- Add interactive slicers for dynamic filtering.
- Enable cross-filtering between dashboard visuals.
- Present healthcare information in a clear and user-friendly layout.

Dashboard KPI Summary Cards

Four KPI cards were created at the top of the dashboard to provide a quick overview of the healthcare data. These cards provide users with an immediate summary of the most important healthcare metrics.

KPI	Purpose
Total Patients	Displays the total number of patient/admission records
Total Billing Amount	Shows the overall billing amount
Average Billing Amount	Shows the average billing amount
Average Length of Stay	Shows the average number of days patients stayed



Demographics & Patient Distribution

The dashboard includes visualizations for analyzing the demographic distribution of patients.

1. Matrix Visual

A **Matrix Visual** was created to analyze patients based on age group, gender, and medical condition.

Matrix Configuration

- **Rows:** Age Group, Gender
- **Columns:** Medical Condition
- **Values:** Total Patients, Average Billing Amount

This visual allows the user to examine patient counts and average billing amounts across different combinations of demographic and medical categories.

Medical Condition	Arthritis		Asthma		Cancer		Diabetes		Hypertension		Obesity		Total	
Age Group	AVG Billing Amount	Total Patients	AVG Billing Amount	Total Patients	AVG Billing Amount	Total Patients	AVG Billing Amount	Total Patients	AVG Billing Amount	Total Patients	AVG Billing Amount	Total Patients	AVG Billing Amount	Total Patients
All	25,436.09	11052												
0-18			25,942.88	150	26,000.60	150	25,672.13	146	26,538.24	140	27,765.62	143	25,480.52	159
19-35			25,383.56	2279	25,314.44	2289	25,411.08	2288	25,541.40	2246	25,658.18	2252	25,342.34	2290
36-50			25,409.92	2405	25,409.44	2310	25,645.60	2405	25,674.87	2400	25,702.44	2403	25,663.03	2405
Total	25,436.09	11052	25,595.81	9308	25,616.38	9185	25,495.10	9227	25,537.11	9304	25,631.92	9245	25,404.10	9231

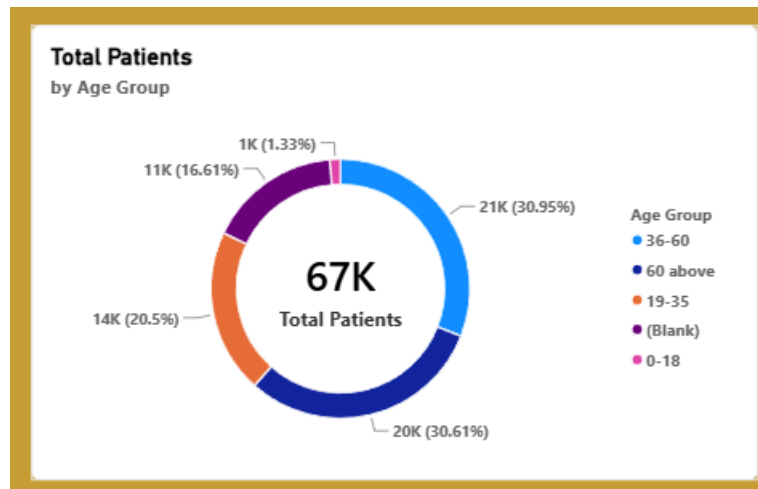
2. Donut Chart

A **Donut Chart** was created to display the distribution of patients across different age groups.

Configuration

- **Legend:** Age Group
- **Values:** Total Patients

The donut chart provides a simple visual representation of the proportion of patients belonging to each age group.



3. Medical Conditions & Admission Volumes

A **Stacked Bar Chart / Matrix** was used to analyze the number of patients associated with different medical conditions and admission types.

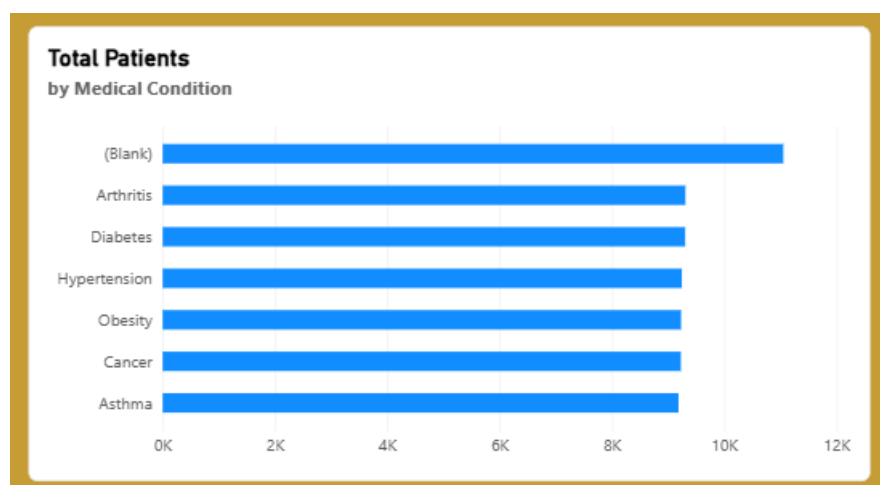
Configuration

- **Y-Axis / Rows:** Medical Condition
- **X-Axis / Values:** Total Patients
- **Split By:** Admission Type

This visualization helps compare patient volumes across different medical conditions while also showing how the admissions are distributed among different admission types.

The admission types include:

- Emergency
- Elective
- Urgent



Admission Volume Trends & Hospital Analytics

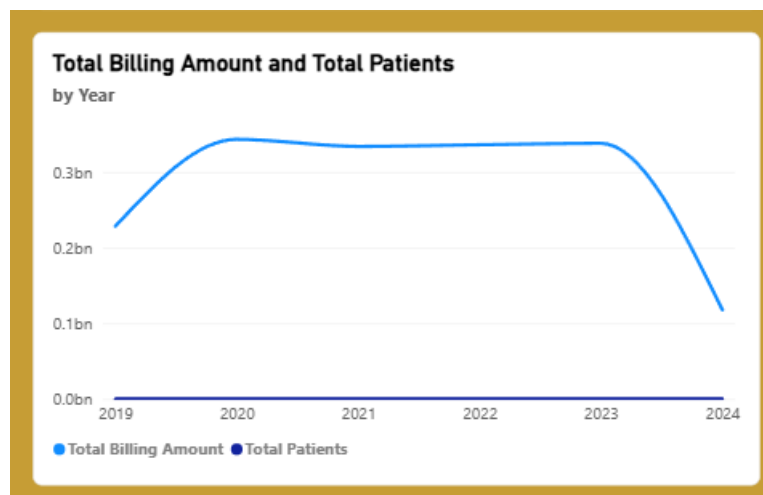
1. Line Chart

A **Line Chart** was created to visualize admission trends over time.

Configuration

- **X-Axis:** Date of Admission — Year and Month
- **Y-Axis:** Total Patients, Total Billing Amount

The line chart allows the healthcare organization to observe changes in patient admissions and billing amounts over different periods. It can help identify changes in admission volume and billing activity over time.



2. Clustered Column Chart

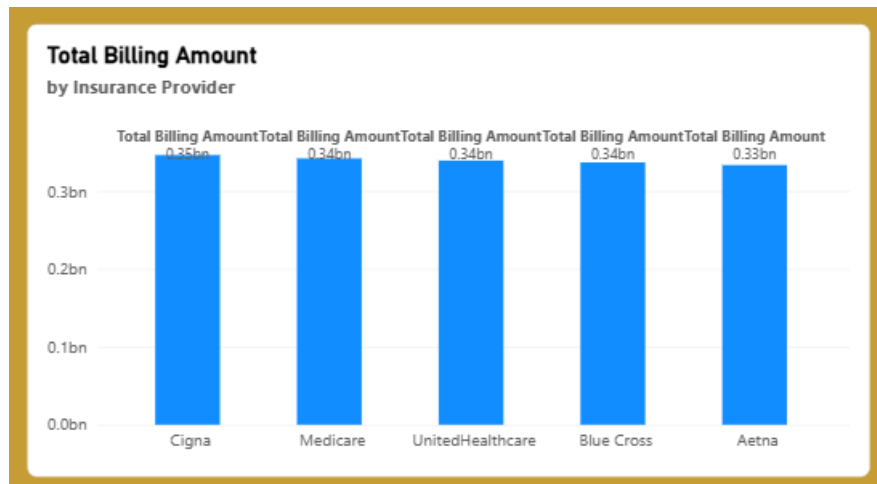
A **Clustered Column Chart** was created to analyze billing amounts by insurance provider.

Configuration

- **X-Axis:** Insurance Provider
- **Y-Axis:** Total Billing Amount

This visualization compares the total billing amounts associated with different insurance providers.

It provides an easy way to examine billing distribution across insurance providers.



3. Slicers & Dynamic Cross-Filtering

Interactive slicers were added to make the dashboard dynamic and user-friendly.

Slicer 1 – Insurance Provider

This allows users to filter the dashboard according to a selected insurance provider.

Slicer 2 – Admission Type

Users can select a specific admission type such as Emergency, Elective, or Urgent.

Slicer 3 – Date Range

This allows users to select a specific period and analyze healthcare information for that date range.

Conclusion

The Week 6 project provided practical experience in **Power BI dashboard design, data visualization, KPI reporting, interactive filtering and cross-filtering**.

Different visualizations were selected according to the type of information being analyzed. KPI cards provide a quick summary, the matrix and donut chart support demographic analysis, the stacked bar chart supports medical-condition and admission analysis, while the line chart and clustered column chart provide time-based and insurance-related insights.

The addition of interactive slicers makes the dashboard dynamic, allowing users to explore the healthcare data based on **insurance provider, admission type, and date range**.

Overall, the completed dashboard provides a clear and interactive way to explore healthcare **patient, admission, and billing information** in Power BI.